

Human senses

THE WAY WE GATHER INFORMATION ABOUT OUR SURROUNDINGS.

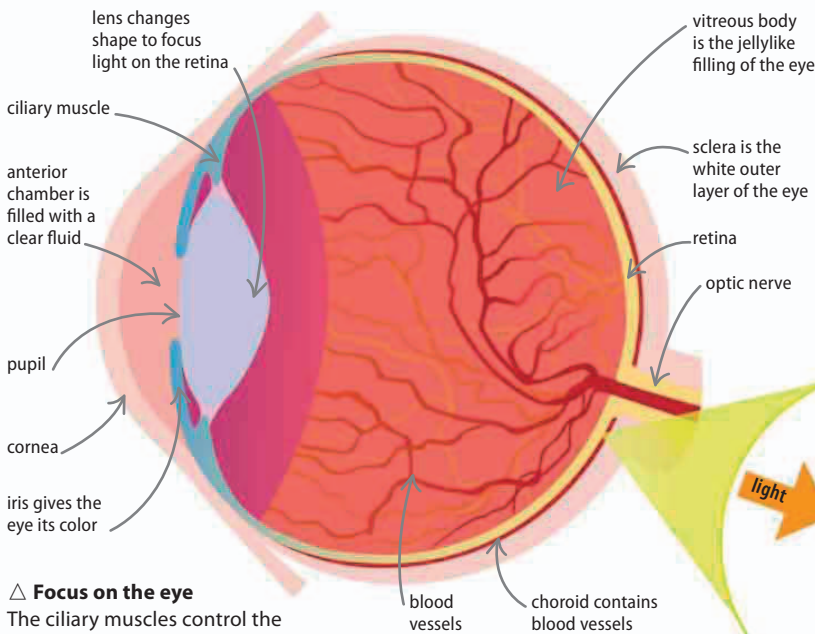
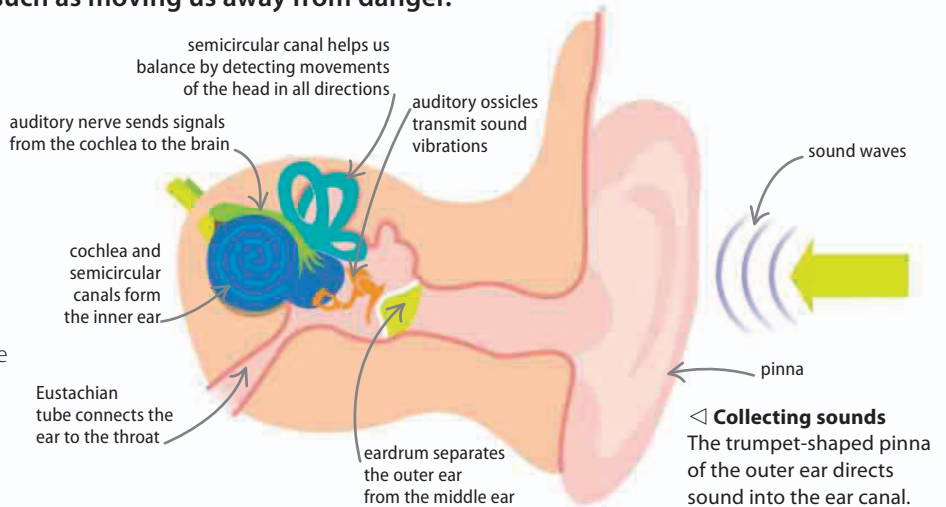
Our senses of hearing, vision, smell, taste, and touch constantly relay information to our brain about the world around us. The brain can then respond if necessary—such as moving us away from danger.

SEE ALSO

◀ 40–41	Sensitivity
◀ 62–63	Body systems
Brain	68 ▶
Optics	198–199 ▶
Sound	200–201 ▶

Hearing

The ear is a hypersensitive touch organ that picks up pressure waves moving through air, which make the eardrum vibrate. This vibration travels along three tiny bones, called the auditory ossicles, to the fluid-filled labyrinth, made up of the cochlea and semicircular canals. The sound waves become ripples in this labyrinth fluid, wafting hairlike sensory nerve endings that send signals on to the brain along the auditory nerve.



△ Focus on the eye

The ciliary muscles control the shape of the lens to focus light from near and far onto the retina. The retina then sends the image to the brain along the optic nerve.

Vision

The eye is like a camera. It lets in light through the pupil, a hole at the front that can adjust its size by contracting and relaxing the iris. The cornea and lens work together to focus the light onto the retina, a lining of light-sensitive cells at the back of the eye. The cells pick up the pattern of light falling across them, which is transmitted to the brain by the optic nerve and made into an image.

◀ Rods and cones

The cells in the retina have light-sensitive pigments that produce electric nerve pulses when light hits them. Rod cells are used in night vision and cannot detect color. There are three types of cone cells, each sensitive to light within a different range of colors, and are used to produce color images by day.

